

MOBILE PATIENTS' TRACKER
(CASE STUDY: KIRATINA HOSPITAL)

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A Research Proposal to The Department of Information Technology in Partial
Fulfilment for the requirement of certificate in information Technology Mount Kenya
University.

MAY ,2023

DECLARATION

DECLARATION BY THE STUDENT

I Hereby that this is my original work based on my creativity and research that has never been submitted in any institution for academic purposes.

Signature.....

Date.....

Reg no: 73990

Name: KIPTOO DOUGLAS WEKESA

DECLARATION BY THE SUPERVISOR

This has been cleared for examination with my approval as the university supervisor.

Signature:

Date:

Name: MRS HARRIET MWENDWA.

Department: Information Technology.

School: Computing and Informatics.

DEDICATION

I dedicate this proposal to my parents who have never failed to give me financial and moral support and for teaching me that even the largest task can be accomplished if it is done one step at a time.

ACKNOWLEDGEMENTS

First and foremost, I'd like to thank Mrs Harriet for instructing us about the methods of conducting feasibility and requirement studies as well as designing. Furthermore, I would like to express my gratitude to thank Head of the institution for the all-time support. Finally, I would like to collectively thank God almighty for his continued blessings, love and protection over this whole process and all people who have helped me in respect to this proposal.

ABSTRACT

This project involves the development of an Android application for viewing and managing patient's medical records. This system is designed for Kiratina Hospital. The application is designed to help doctors view the medical history of their patients. The doctor can use the application to get all the medical history of a patient at any given point of time. The doctor can also check on all the patients he treated for any day of the calendar. Doctors can log in the app from their smartphones and enter relevant information about their patients, including patients name, pending appointments, disease history, medication administered, date of arrival, and consultation and treatment costs among other things. Doctors can also continually upgrade the system by entering the latest medical data of their patients. The doctor can also check on all the patients he treated for any day of the calendar. Currently Kiratina Hospital is using paper-based way of recording medical data, one of the biggest drawbacks of paper-based registration systems is the associated costs. In addition to tones of paper, you will need more printers, photocopier, stationery, and other office supplies. These costs add up and can become a significant expense in Kiratina Hospital also which is time consuming during data entry, needs larger storage location for filing, higher risk of damage and there is risk of being accessed by unauthorized personels and long queue causing congestion as a lot of time is wasted in registering a single patient. The main objective of this research is to design an online patient tracking system that will deal with the information provided by patients to help doctors run their work efficiently, saves time by reducing paper work and long consultations also protecting patients confidentiality. Having a patient's medical and contact information readily available can be potentially lifesaving during critical medical events such as severe allergic reactions or heart attacks. Moreover, by reducing errors and saving time, MPT may therefore help reduce the large number of deaths attributed to medical errors. Once the information is saved on the system, it will be stored for the perusal of doctors for their future cases. So, if a patient comes for a follow up routine, doctors can immediately check up on their medical history and provide the necessary and relevant medical treatments. HTML5 and java script programming languages and MYSQL will be used for the development of the GUI and SQLite for the development of the database and ECLIPSE for development environment. The design tools that will be used include data flow diagram and flowchart. The development of this system will help in analysing and generating Information, allowing doctors to view the status of the information they have lodged. The system will be accessed using the correct username and password. This application can be installed on any Android device.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the study

A system is an array of components designed to accomplish a particular objective according to plan (Rosenzweig 2014). A management system according to Wikipedia is a framework of policies, processes and procedures used by organizations to ensure that it can fulfil the tasks required to achieve its objectives. Mobile patient tracker is an application designed to help doctors to see the medical history of their patients, see pending appointments and manage patients profiles. The application allows doctor to insert various data fields regarding a patient including patient contact information, age, blood group, gender, medical provided, date of visit which the system automatically records. Most traditional information systems have been developed for the latter and meet their needs well but may have limited relevance to the busy workers.

Kiratina Hospital is a medical clinic located in Nakuru county in Nakuru North District. This medical clinic is owned by Private practice-Nurse/midwife and it plays a big role in Rift valley province, which runs on 24/7 fulltime. Kiratina Hospital sometimes have large number of patients coming in for different services. In most cases some patients go back home unattended to due to congestion causing busy schedule unabling the doctors to attend to patients at the right time. There are several methods that are used to attend to new or/and regular patients . Most traditional information systems have been developed for the latter and meet their needs well but may have limited relevance to the busy workers.

Doctors can log in the app from their smartphones and enter relevant information about their patients, including patient's name, disease history, medication administered, date of arrival, and consultation and treatment costs among other things. Doctors can also continually upgrade the system by entering the latest medical information of their patients. One of the methods is through interviewing patients one by one and filling their info in paperwork. The problem with this method is that a lot of time is wasted in interviewing the regular patients since there are no

records for previous or past visits hence the long queues in the organization during workdays which can lower patients risk of getting hospital acquired information. Secondly, in an event where a lot of patients need to be served the hospital tend to verbally interview the patients. The biggest problem with this method is majority of the patients fear airing their information in a room full of other patients for fear of victimization in case of a confidential problem/illness, secondly information are not preserved for future reference which may lead to wrong diagnosis of patients in the hospital. In most cases a lot of information is lost hence the need to reinterview the patients. Thirdly one of the biggest drawbacks of paper-based registration systems is the associated costs. In addition to tones of paper, you will need more printers, photocopier, stationery, and other office supplies. These costs add up and can become a significant expense in Kiratina Hospital. Many of today's employees want to work for businesses that prioritize sustainability, and many consumers also prefer environmentally friendly alternatives, so you do your best to attract and retain them.

1.2 Problem Statement

Kiratina Hospital uses the hands-on process to provide health services, use of card, files to track patient progressive records, and set records of new patients in the hospital. This procedure has a lot of inconsistencies such as patients losing their health cards, or their files misplaced. The use of verbal interview poses fear of patient's confidentiality, congestion in the hospital as more patients come in before others are attended to, very costly to some patients and high-risk data loss in case of fire or natural disaster. On the other hand, the nurses or doctors consulting a patient verbally may end up forgetting or losing patients information's in paperwork form. Also, use of paper-based registration system takes up a lot of space and can contribute to an untidy environment. This program improves the quality of service in terms of assistance provided by empowering physicians and administrators to be able to view the level of common diseases and their percentage. Doctors can also determine the number of patient visits over a set period.

1.3 Proposed solution

The proposed solution is to develop this android mobile application system that will enable the organization to access their client's information anytime with ease, register new patients with a click of a button, with little or no paperwork hence reducing congestion and saving a lot of time to attend to other patients.

Doctors can easily add new patient details and track the patient's history at any time all on their android phones. This system will keep record of all patients offered for future reference and doctors can edit any information on any patient at any time when needed. This system is environmental conscious, meaning fewer paper files means less paper used for duplicate copies and for general record keeping. Also, the application will help improve bed allocation, upgrade security, and increase patients' satisfaction. The system will also keep track of time when the doctor logged in to a patient's profile, patients time in and time out of the visitation. Also, the system will remind doctors of pending appointments with different patients sorted by time and enable doctors to reschedule the appointment timetable if needed. If an appointment is rescheduled the system will automatically send a notification to the patient notifying them of the new set date and time.

Basic sensor readings can be collected from the mobile phone itself like the heart rate, and whenever there is some abnormal readings, the application warns the patient and after some threshold informs the hospital. It is an easy-to-use application that is easily operatable to users.

1.4 Objectives

1.4.1. Main Objectives

The main objective of this research is to design a mobile patient tracker that will help Kiratina Hospital to deal with their patients in an orderly manner, making sure no time is wasted, information is recorded permanently, and patients are acted upon immediately and eliminate the drawbacks in the system such as lack of proper records and thus help in reducing congestion in the Hospital organization.

1.4.2 Specific objectives:

- i. To develop a system that will enable Kiratina Hospital to register patients online for easy storage of patient medical data to avoid congestion
- ii. To develop a mobile application system that will send a reminder to the patient on pending doctor's appointment.
- iii. To develop a mobile patient tracker that will generate comprehensive health record as a guide in giving them proper diagnosis
- iv. To develop a mobile application that will generate the patients pending hospital bills and any outstanding balances.

1.5 Research questions

- i. How will the system register patients online in Kiratina Hospital?
- ii. How will the system application send a reminder to the patients on pending and new appointments?
- iii. How will the system generate comprehensive health records?
- iv. How will the application generate patients pending bills and outstanding balances?

1.6 Project scope

This project will be designed for Kiratina Hospital, but it can be used by other hospitals that would wish to improve on their patients' services and improve organization image.

The project will have the following modules: **The registration module**- which records all the personal details of the patients and enter them in the Hospital database system. **The login**

module-which allows Kiratina doctors to log in the system and lodge patients' information and log out of the system. **The report module**- which the doctor can go through and correct any error made during the registration process. Scope of proposed system is justifiable because large amount of the population faces the problem of managing the medical records in form of files and papers.

1.7 Project Justifications

Kiratina Hospital faces problems that can be solve using technology. This application will help to manage some of the problems and help the organization execute better responsive mechanisms to patients. In the process it will help stem most of the patient's unrest that result from poor services in the organization. Also, the application will help to upgrade security in the hospital. This application will use internet technology to allow doctors to lodge data quickly and efficiently.

1.8 Project risk and mitigation

Every system comes with a measure of risks and dangers. The mobile patient tracker system also comes with some risks and these risks are:

Cybersecurity risks: Risk related to breach of protected health information of patients. The system is vulnerable to hacking, which means sensitive patient data could fall into wrong hands. To curb this problem, the system will be secured with data encryption software, passwords, and firewall. Confidentiality of data: the extent to which the ability of the organization to provide a service will be affected by the disclosure of the given set of data to an unauthorized person. Application usability: some of the doctors may not have knowledge to use the system. It also requires time for the hospital to adapt fast. There is problem of over-dependency on the system which may slow services when the system is down.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction to literature review

A literature review is a search and evaluation of the available literature in your given subject or chosen topic area. It documents the state of the art with respect to the subject or topic you are writing about. (Lizard, 2018). This review documents, similar projects that have been done regarding mobile patient tracker registration, tracker analysis, electronic medical records, online registration, and reminder sending on pending appointment to the appropriate patient. The huge growth of information in the medical environment deemed the necessity of an information system. Both clinical and related medical information that managed by this system.

2.2 Patient tracker online registration

According to (Rahman, 2014) the system was implemented to work on the concept of registering patients digitally which is the replacement of existing paper registration system which is a complicated method for collecting data from patients. In the sample system, three main forms were designed, i.e., doctor login form, admin login form and registration menu.

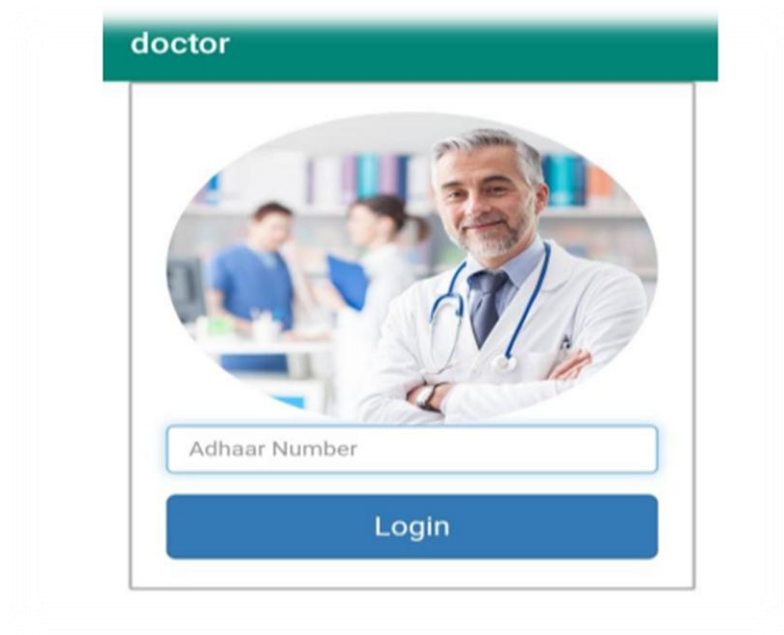
Doctors will login with a valid username and password if he/she doesn't have a profile registered s/he has to go to the registration page and register as a doctor. If the doctor wants any changes to their profile, then there's an option for viewing and editing information. Admin will find all the patients' information, add, edit and/or generate a patient's report. After that logout button occurs and admin simply logout from the system. There also the registration menu, which is the main feature in the application. This page has all the fields the doctor must feed in about a specific patient.

Fig 1.0 Sample Design of online registration



The image shows a sample design for an online registration form for a doctor. At the top, there is a green header bar with the word "doctor" in white. Below the header, there is a circular profile picture of a doctor. Underneath the profile picture, there are two input fields: "Username" and "Password". Below these fields, there is a blue button labeled "Login". At the bottom, there is another blue button labeled "Create New Account".

Source (kenedy,2017)



The image shows a sample design for an online registration form for a doctor, similar to the one above but with a different input field. At the top, there is a green header bar with the word "doctor" in white. Below the header, there is a circular profile picture of a doctor. Underneath the profile picture, there is a single input field labeled "Adhaar Number". Below this field, there is a blue button labeled "Login".

Source (kenedy,2017)

2.3. Appointment and Message alert system

The first message was sent in 1992 from Neil Papworth, a former developer at Sema group Telecoms. Mobile didn't have a keyboard at that time, so Papworth had to type the messages on a PC. This will help the company to notify the patients on pending appointments and payments. The doctor sent message to a patient to alert them that their registration and transactions had been successfully received also If the doctor is available for appointment.

Fig 3.0: add appointment or/and add patient



Source (Nicholas,2015)

2.3.3. Health records generator system

This system option collects all the vital health status of a patient and generate a comprehensive health record which includes blood type, age, allergies etc. The doctor uses this report to diagnosis the patient with proper and effective drugs.

Figure.3.2 Health records generator system

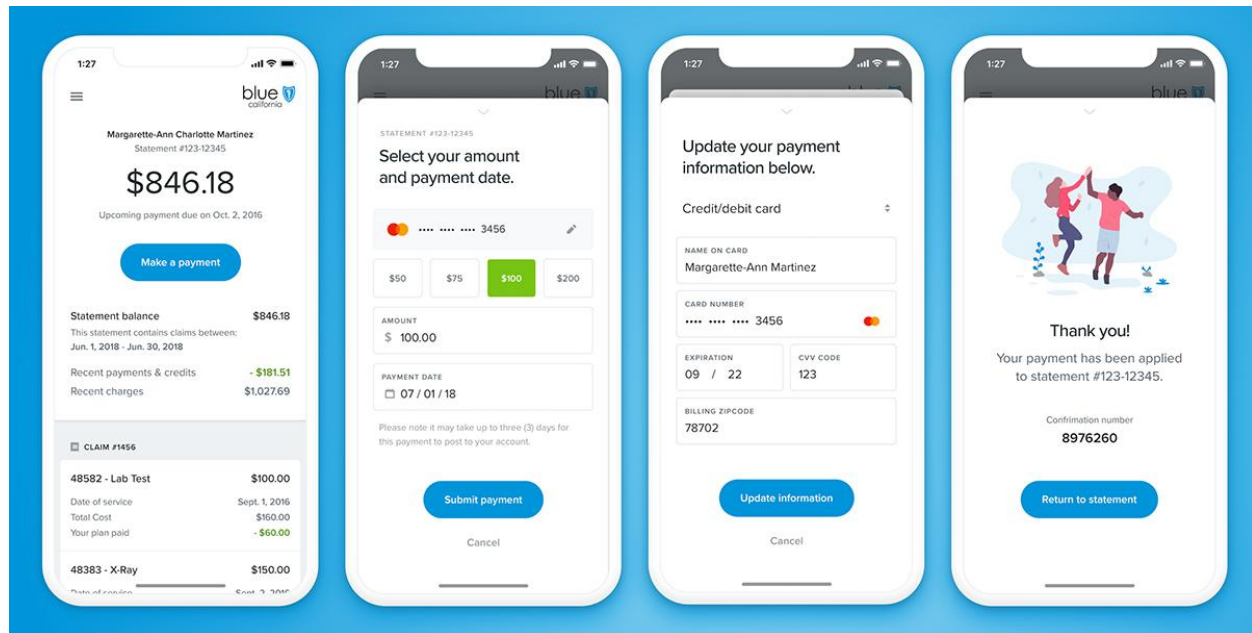


(Source. Finlay's 2016)

2.3.4. Payment system

This is a process whereby a patient pays for services and medication offered by the Hospital in which the doctor provides a billing report for a patient at the end of treatment section. This is conducted in a proper manner to ensure proper documentation incase the patient is covered by an insurance. The patient can later proceed with the bill's payment through the hospital banking account. As the world advances more with technology development, there is a rise of electronic payment systems and payment processing devices.

Figure.3.3 Payment method



(Source. Richard ,2016)

2.4 Design

The design stage has been defined by the route of the patient from entry to the hospital until departure. This is necessary so that all systems can provide data relating to the patients care attendance in the prescription-validation-preparation/dosage/dispensation-administration process of medication in Kiratina hospital. It was decided to use ECLIPSE Programming environment for development as one of the most sufficient and user friendly. SQLite shall be used to develop the system database. This is needed because it is not necessary that there will always be a network to send the biometrical data to the server. Android SDK: To create the android application. Android is a Linux based operating system for mobile devices smart phones. It is developed by the open handset alliance led by Google. The android open source (AOSP) is tasked with maintenance and further development of Android. JAVA is used in a wide variety of computing platforms from embedded devices and mobile phones on the low end to enterprise servers. Java is used to provide improved and secure functions while browsing the internet. ECLIPSE and XAMPP. Android Studio used for Android application development.

2.5 Research Gap

From the research, it is evident that the employees aid groups faced tremendous logistical challenges in using the existing patient's management system. They also needed to expand their IT staff and conduct trainings to both employees that are supposed to use the system. There is limited scientific research on this system in developing countries. Kiratina Hospital also had a lot of paperwork since most of the patient's data were recorded in paper.

2.6 Conclusion

The proposed application is an easy-to-use application that helps Kiratina Hospital store their patient's data on a server and serve them effectively and equality. Can be accessed anytime by the relevant users. It keeps patients and doctor records on one platform it is therefore easily reliable and efficient. This app is designed for a short time has many features that suits the user according to their requirement. The app requires less space and is more powerful compared to other existing applications. There is an added range of improvements to this application for user requirements. More features can be added to this app in the future.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

According to (Ballenger, 2015) a research methodology is the process used to collect information and data with the purpose of deciding. The methodologies include interviews, observations, and questionnaires. Also, Methodology refers to the analysis of the methods used appropriately to a field of study. It is a systematic way of accomplishing certain task and is a collection of procedures, techniques, tools, and documentation aids that help system developers to speed up and simplify the system development process (Tylor, 2001).

3.2 Methods of data collection

To collect the required data analysis, the following research methods will be adopted: Observation, Questionnaires, Interviews, and document review.

3.2.1. Observation

The hospital management team in Kiratina will revisit and check the daily routine in the Hospital environment and daily operations. The team closely observes the organization activities and process taken by patients from the time they enter the hospital premises, to how they are attended to by the hospital staff, up to the time they leave the hospital. This is a simple method that will make the researcher to observe the activities taking place without disrupting normal operations.

3.2.2. Questionnaire

These are forms which are completed and returned by respondents within the hospital organization who are given questions randomly to answer. The researcher will use both the open-ended and closed ended questionnaires in gathering information to determine the efficiencies and inefficiencies of the existing system.

3.2.3. Interviews

Interview will be carried out within Kiratina Hospital so that the researcher will gather information about the existing system in terms of service delivery and efficiency of the system. The advantage of using interviews is that it is cheap and firsthand information is collected from the interviewees.

3.2.4. Document review

The researcher will explore existing records to provide insights into the organization regarding issues that will not be covered in the previous research methods. The researcher will be required to ask permission from the Hospital administration to review the Hospital documents.

3.3. Feasibility Study

Preliminary investigations examine system feasibility: the likelihood the system will be useful to the organization, the main objective of the feasibility study is to test the Technical, operational, and Economical feasibility for adding new modules and debugging old running system.

3.3.1. Technical feasibility

The feasibility will check whether the existing technological resources and manpower can help operate on the new system and if not what extra resources are required so as resource system. These resources are needed to meet the user requirements.

3.3.2. Economic feasibility

Economic feasibility is concerned with checking whether the proposed system will be costly than the existing one. The cost incurred in development of the system should be well compensated by the system.

3.3.3. Operational Feasibility

Operational feasibility was concerned whether the new system would be useful upon development and its implementation. It also checked whether the patients would be comfortable with new system and if it reduced time wastage and help ease the service.

CHAPTER FOUR

SYSTEM ANALYSIS

3.4 Introduction

The analysis stage focuses on determining which information is most relevant for study to determine the pros and cons of the system and which professionals will provide more data in respect to each of the indicators following their experience of the system in daily clinical practice.

3.4.1. System Requirements

An android smartphone from version 4.4(Kit Kat) up to date will be used due to its usability and ease to operate, consume less power, more affordable and can be updated to support the latest version of the system.

32 GB Memory Card will be used to increase data storage capacity and create space for data backup and development of the system.

Epson L655 will be used for printing any required document e.g., the bill payments and insurance section.

3.5. System Design

The doctor logs into the system with correct username and password. The doctor is then verified by admin. The doctor registers a patient on the system with detailed information, after registration each patient registered can view their profiles. The patient can fill in their report or have the doctor fill for them before reviewing and confirming. The doctor/nurse can then search patients' information by name/ID or date of registration. Before administering any medication to a patient, the admin must generate health records for proper diagnosis. After patients' diagnosis the admin can then generate pending bills. The admin checks or set appointments when necessary. From there the doctor/nurses can manage different patient's profile. The admin can later backup the medical data into the database.

3.5.1. Logical design

The design will lay out the system components and their relationship to each other.

Fig 4.0

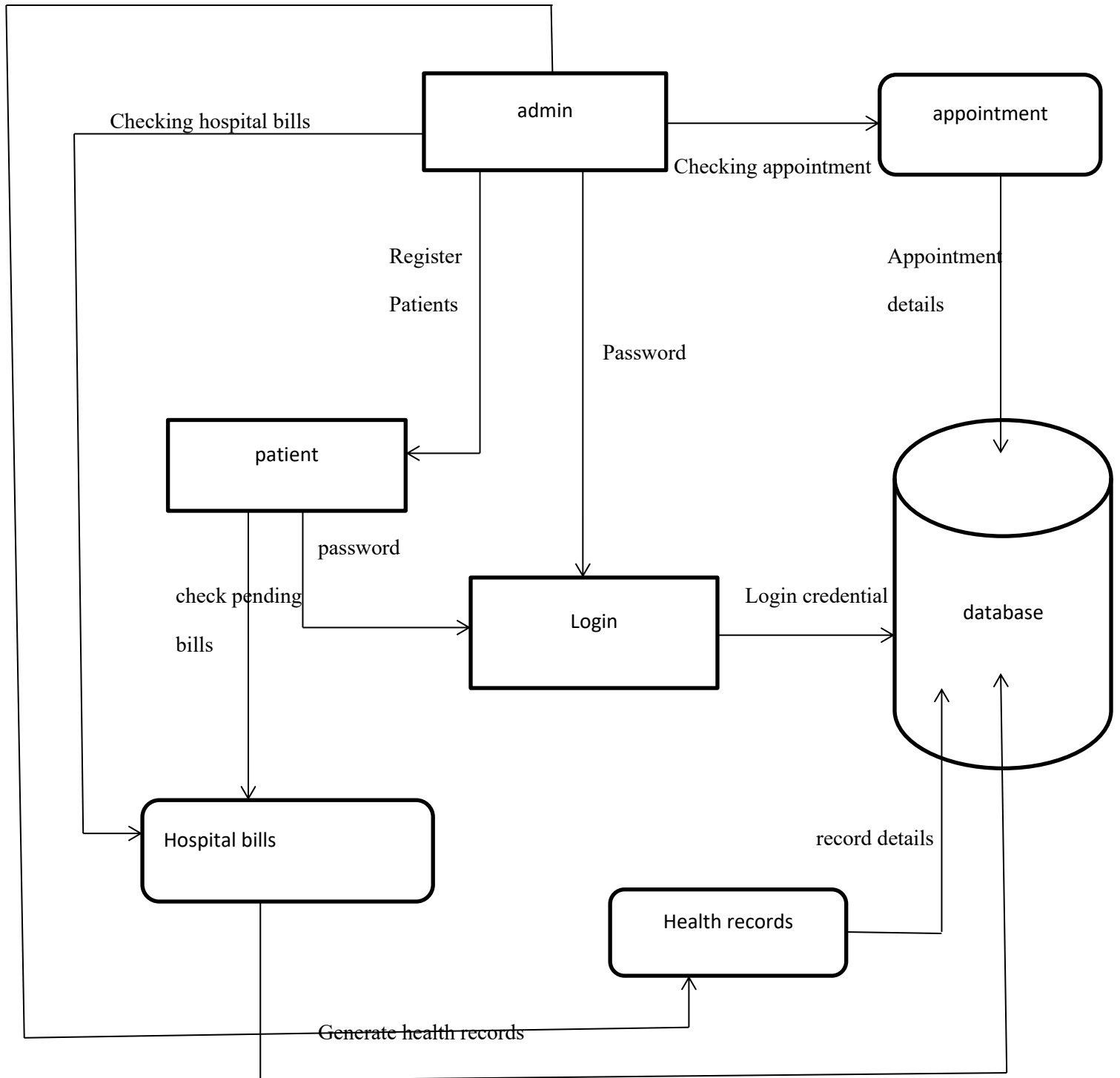
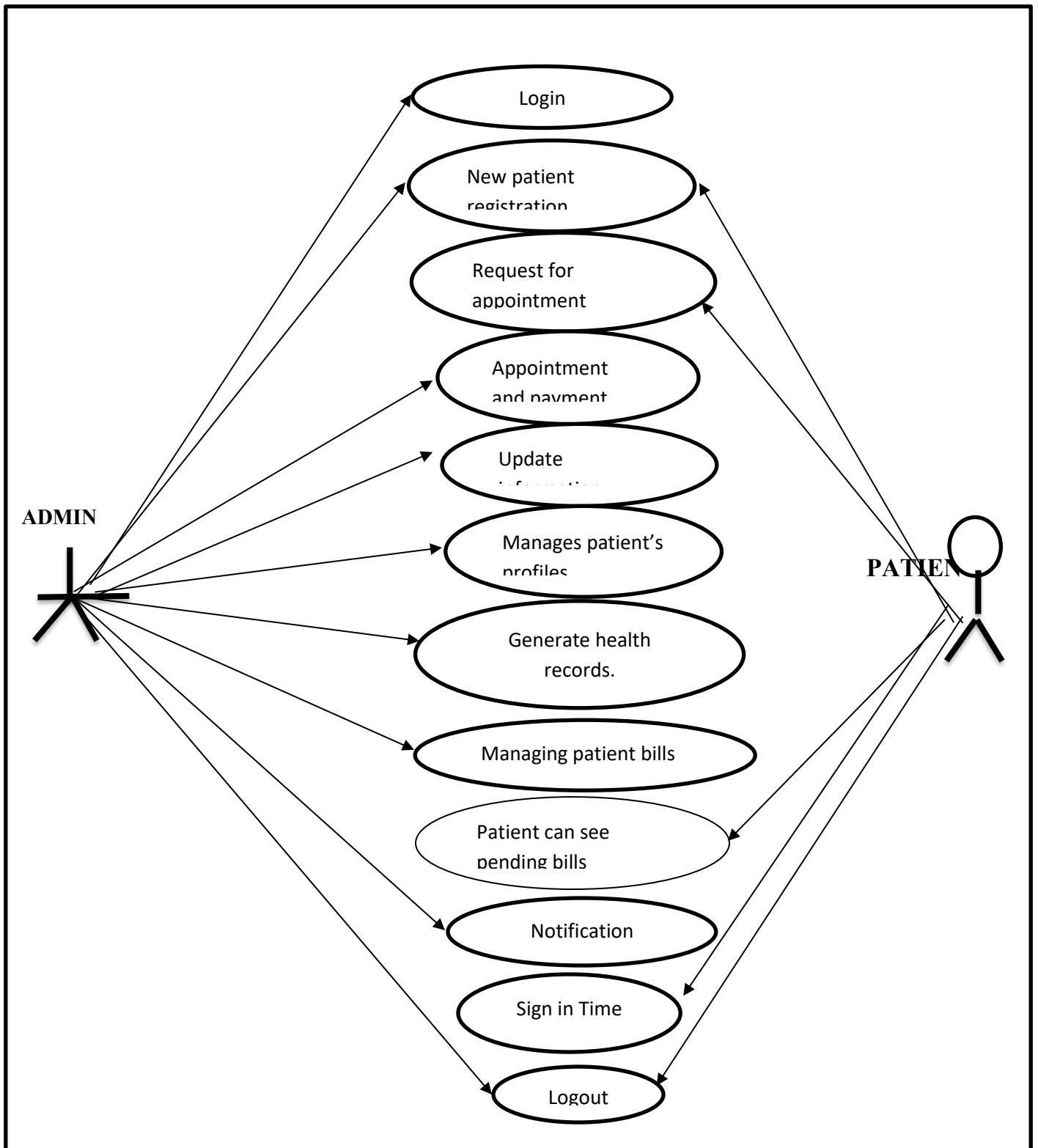
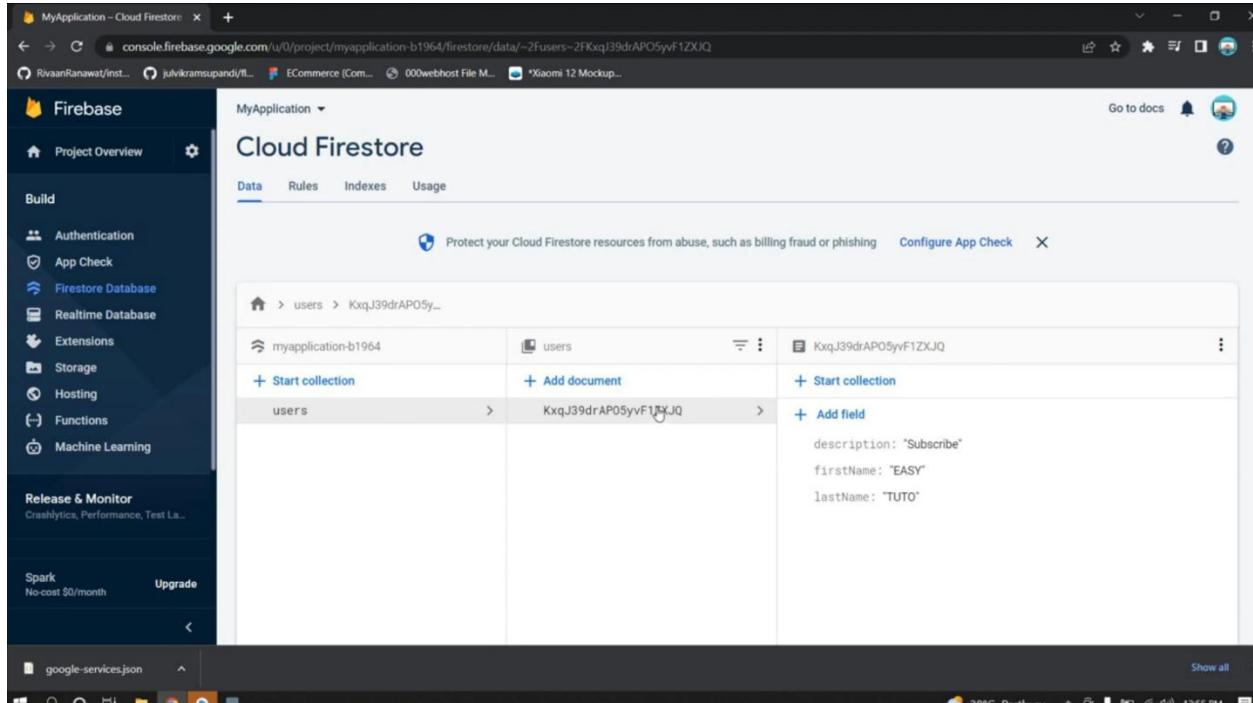


Fig 5.0. Use Case Diagram



Database design

Firebase is Google's mobile application development platform that helps us to improve, build and grow our application. We are using its real-time database feature to store the data of users like password and login credentials in a data base.



We will be creating a data base, a real-time data base which is a feature of Google firebase to store all the data of users which will be their ID Card Number. Secondly their usernames and passwords and lastly all the data which will be retrieved from their ID Card like their Phone Number, Address, Pin Code but for that all the users must register on the application first.

CHAPTER FIVE

SYSTEM DESIGN

Introduction

The system design chapter helps in producing the design structure of the new system. It explains how the information flows on the new system, the interactions between the modules and the database structure of the system.

System design also explains the logical layout of the new system. It explains how input of data, processing and output of the desired information is achieved successfully. It consists of all the system modules, database layout and the procedures followed by the production procedures.

Description of System Modules

CHAPTER SIX

SYSTEM IMPLEMENTATION

Introduction

This chapter explains how the system will be put into action after it has been developed. This will ensure that the current manual system is replaced, and it will run smoothly. It will give different implementation strategies, and post implementation review is also discussed.

System integration

This is whereby the individual components of the software are combined without any prior testing and then tested as a complete system. Through incorporating all the modules into one system. This made all the system modules work together as one and was done to ensure that the system work effectively and successfully with every module relating well with others by ensuring the system doesn't have any errors.

System testing

The system testing is critical element of software quality assurance and represents the ultimate reuse of specifications. A strategy for software testing must accommodate low-level tests that necessary to verify that a small source code segment has been correctly implemented as well as high level against customer requirements.

Validation Testing

The test aims at ensuring correctness of the system. It ensures that all unit's programs conform to each other without conflicting. Security will also be an issue thus a username and password are required for logging in.

System implementation

The proposed system will be implemented in parallel to the existing system. This is to ensure that the users of the system will interact with the system and get to know its dynamics. Also, in parallel implementation there is little risk of data loss because the known-good system is running.

System Maintenance

The proposed system will be maintained through corrective, adaptive and perfective maintenance methods. Corrective maintenance will error in the proposed system that could affect various areas of the system, design, logic, or code. Most commonly, these changes are sprung by *bug reports* created by users. Most of the time, the problem reports submitted by the users may serve and enhance the system. Adaptive maintenance is triggered by changes in the environment the system lives in, for example by the changes to the operating system, hardware, software dependencies and even organizational business rules and policies. Perfective maintenance will come in to modify or enhance the system according to the users' requirements and changing needs.

Unit testing

This test ensures that the application produces the desired results. This will test the aspect of each form by using valid and invalid test data. The test data will be added to test code associated with the module and results compared with the expected results. Logical testing will aim at producing an application that produces the required results that are error free.

Validation testing

The test aims at ensuring correctness of the system. It ensures that all the integrated units programs conform to each other without conflicting. Security will also be an issue thus a username and password are required for logging in.

Integration testing

This test is carried out when the entire system is complete, and all the modules have been tested and each module successfully performs its functions. The whole system is tested to ensure no errors have been introduced. It involves controls such as command buttons to perform operations such as adding new entries, saving, editing, deleting an entry, and searching the database.

System implementation

This is concerned with the smooth shift from one way of doing things to another and how the disruption of business activities will be mitigated during the shift process. There are three main methods used for changeover: Direct changeover, parallel running, and phase implementation.

System implementation was carried out by use of direct changeover that ensured smooth shift from paper-based mode publication to that of online based noticeboard.

Direct changeover

The method of direct changeover as a means of system implementation of the new system was used because of the benefits that came along with it. The benefits that lead to the use of direct changeover as a means of implementation includes less time slot, the method is less costly as compared to other methods like parallel running, the method is less risky. This method in case of any challenges that may arise on its use, the return to the old system that is paper based management publication can be achieved within the shortest time possible and in a less expensive way.

CHAPTER SEVEN

LIMITATIONS, CONCLUTIONS AND RECCOMENDATIONS

Limitations

The following drawbacks were encountered when coming up with this research project.

Limited scope of testing: Testing will only be carried out in the environment it exists for example, testing in an Android 5.0 Lollipop and above system only.

Some errors may be untraceable. Not all the required tests may be accomplished therefore the system will contain few problems in its operations. There will be bugs in start, which will need time to fix.

Conclusion

The online poultry system will help farmers who wish to use this system to manage their poultry produce. The system will also store data that will be used for future reference in poultry management.

Recommendation

It would be recommended that stuff and farmers to be well educated on how to use this system to improve efficiency.

